Flora 299 (2023) 152198

Contents lists available at ScienceDirect

Flora

journal homepage: www.elsevier.com/locate/flora

Flower size affects bee species visitation pattern on flowers with poricidal anthers across pollination studies

Tamiris Delgado [a,b,*], Laura Carolina Leal [c], Juliana Hanna Leite El Ottra [d], Vinicius Lourenço Garcia Brito [e], Anselmo Nogueira [b]

[a] Programa de Graduação em Ciências Biológicas (Botânica), Instituto de Biociências, UNESP – Universidade Estadual Paulista, Botucatu, SP, Brazil
[b] Laboratório de Interações Planta-Animal (LIPA), Centro de Ciências Naturais e Humanas, Universidade Federal do ABC, São Bernardo do Campo, SP, Brazil
[c] Departamento de Ecologia e Biologia Evolutiva – Universidade Federal de São Paulo, Diadema, SP, Brazil
[d] Departamento de Botânica, Instituto de Biociências, Universidade de São Paulo, São Paulo, SP, Brazil
[e] Instituto de Biologia, Universidade Federal de Uberlândia, Uberlândia, MG, Brazil

ARTICLE INFO

Edited by: Timotheus van der Niet

Keywords:
Flower evolution
Buzz-pollination
Bee diversity
Flower attraction
Pollen flower
Quantitative review

ABSTRACT

Flower size is a crucial trait for pollinator attraction. In flowers with poricidal anthers, the plant constraints the pollinator's access to the flower's rewards by enclosing pollen grains in poricidal anthers. The pollen is only released by mechanical vibrations applied mainly by bee species during buzz pollination. Although buzz-pollinated flowers exhibit a large size variation across species, their influence over patterns of flower visitation by different pollinator species remains to be elucidated. To fill such gap, we asked how flower size influences flower visitor assemblage, especially bees, across plant species and how flower size is related to body size and vibrating behavior of the most frequent bee species. We hypothesized that the assemblage of animals, especially bees, visiting larger flowers have higher taxonomic and functional diversity than smaller flowers, and the most frequent bee species visiting larger flowers have larger body size and a mandatory vibrating behavior. To test this hypothesis, we conducted a systematic review of pollination studies that reported patterns of flower visitation in plant species with poricidal anthers, describing the relationship between flower size and flower visitor metrics. We found that large-flowered species were visited by three times more species than small-flowered species, including vibrating and non-vibrating bee species. Unlike our expectations, non-vibrating bees were largely absent from small-flowered species. The most frequent bee species visiting flowers, irrespective of flower size, exhibited vibrating behavior. The most frequent bee species visiting large flowers were twice the body size of those visiting small flowers, highlighting a morphological match between flowers and pollinators. Although larger flowers are more susceptible to interactions with exploiters, reproductive success should be favored owing to the higher probability of their interaction with vibrating, large-bodied bee species. Still, a reduction in flower size probably prevents interactions with non-vibrating bees, while favoring vibrating small-bodied bee species.

1. Introduction

Flowers bear different synorganized structures that can protect gametes and flower rewards from antagonists, but attract potential pollinators (Armbruster, 2014; Diggle, 2014). The corolla, which can vary in size, shape, symmetry, color, and smell, commonly performs pollinator attraction (Fenster et al., 2006; Goulson, 1999). Other flower structures can also perform pollinator attraction, such as the calyx or the androecium (Bai et al., 2011; Gumbert et al., 1999). These organs alone, or together, can provide different cues indicating the location and quality of a pollinator's rewards, thus influencing the chances of flower visitation and, ultimately, effective pollination (Latty and Trueblood, 2020). Since most pollinators are visually oriented, pollination attraction is often based on the visual stimulus that also provides cues about flower rewards to pollinators (Chittka and Raine, 2006). Thus, a flower's visual attractiveness can indicate how rewarding a flower can be to a pollinator.

Flower size is positively associated with visual attractiveness to pollinators since it indicates the quantity of potential reward (Fenster et al., 2006), even though the flowers of some species are concatenated

* Corresponding author.
E-mail address: tamiris.delgado@unesp.br (T. Delgado).

https://doi.org/10.1016/j.flora.2022.152198
Received 6 June 2022; Received in revised form 9 November 2022; Accepted 23 November 2022
Available online 30 November 2022
0367-2530/© 2022 Published by Elsevier GmbH.

in inflorescence (e.g., Fisbein and Venable, 1996), so the unit of attraction is not always a single flower. Overall, plant species with large flowers are more attractive to pollinators, receiving higher visitation rates than plant species with smaller flowers (Eckhart, 1999). Numerous studies report this positive relationship between floral size and visitation rate, linking this increase in flower visitation with higher attractiveness and resources available in relatively large flowers (Ashman and Stanton, 1991; Campbell et al., 1991; Fausto et al., 2001). For instance, experimental studies have verified the preference of bees for artificial flowers with a higher nectar quantity commonly linked to flower size (e.g., Blarer et al., 2002). Also, a study with *Rosmarinus officinali* found that larger flowers were more attractive and offered more resources for bees than smaller flowers (Herrera, 2005). Therefore, it is expected that larger-flowered populations have higher visitation rates than smaller-flowered populations in the wild (e.g., Elle and Carney, 2003). Flower size associated with flower resources can play an essential role in pollen removal and deposition patterns by pollinators owing to its influence on pollinator attraction (Rademaker et al., 1997).

The relationship between flower size and pollinator attraction has been mostly recorded and quantitatively revised in flowers that produce nectar as the primary reward to pollinators (e.g., Ashman and Stanton, 1991; Elle and Carney, 2003; Fausto et al., 2001). However, this relationship is still poorly understood in plant species offering pollen grains as the primary reward to pollinators (hereinafter termed as poricidal flowers – Buchmann, 1983). Poricidal flowers are estimated to comprise 6 to 8% of all angiosperm species (Buchmann, 1983), including plant species bearing (i) traditional pollen flowers, defined as "flowers which offer a surplus of pollen as the only reward to their insect visitors, instead of nectar or other substances" (Vogel, 1978), and (ii) poricidal flowers that produce additional flower resources, such as nectar or oil (e.g., some Melastomataceae species; Brito et al., 2016; Kriebel and Zumbado, 2014). In these species, the consumption of pollen grains by flower visitors can directly compromise the pollination process and the reproductive outcome of these plants (De Luca and Vallejo-Marín, 2013; Vallejo-Marín et al., 2009). On the other hand, enclosing pollen grains in poricidal dehiscent anthers is an effective strategy for reducing conflict between flower visitors and plants by restricting the access of flower rewards to a narrower set of visitors able to vibrate the anthers and extract the pollen (i.e., buzz-pollination) (Goulson, 1999). These buzz-pollinators are mainly vibrating bee species that apply mechanical vibration indirectly with their flight muscles to poricidal anthers (De Luca and Vallejo-Marín, 2013; Vallejo-Marín, 2019). As a mechanical response to the bee's vibration, pollen grains are mostly immediately released from the flower through the terminal anther pores. Then, some pollen grains are actively captured by bees, while other pollen grains are dispersed on their bodies (Buchmann, 1983; Hrncir et al., 2008; Michener, 1962). Since poricidal flowers vary in flower size, reward quantity and visiting pollinator species, especially the bee assemblage, the relationship between such variables may be similar to that observed on nectar-producing flowers.

In *Solanum* species, small-flowered plants with poricidal anthers have reduced floral organs and resources, with lower spatial segregation between anthers and stigma and a lower pollen-ovule ratio than large-flowered species (Vallejo-Marín et al., 2014). Consequently, it has been suggested that smaller poricidal flowers, like nectar-producing species, are less attractive to pollinators and consequently attract a smaller bee assemblage than larger poricidal flowers (Vallejo-Marín et al., 2014). This pattern was also observed in Melastomataceae, in which species with larger flowers had a greater richness of visiting bees (Valadão-Mendes et al., 2022). An exciting aspect emerging from studies with poricidal flowers, in contrast to nectar-producing flowers, is the potential relationship between flower size and functional traits, especially vibrational features, of bee assemblages visiting such flowers. For instance, smaller- and larger-flowered poricidal species are more often visited by vibrating than non-vibrating bee species (Brito and Sazima, 2012; Souza et al., 2012; Valadão-Mendes et al., 2022), but more rarely by other functional groups (e.g., Fidalgo et al., 2018; Quakenbush, 2018; Souza et al., 2012). Furthermore, smaller poricidal Solanaceae flowers frequently interact with small-bodied bee species (Vallejo-Marín et al., 2014), which exhibit a lower ability for pollen extraction in comparison to the large-bodied bees (Stout, 2000). Also, bee species that cannot vibrate the poricidal anthers (e.g., *Trigona*) are often found on larger-flowered species (Figueiredo et al., 2009; Nogueira et al., 2018). This pattern is probably recurrent, given the greater number of resources in larger flowers, which allows a greater assemblage of visiting bees, including non-vibrating bees as antagonists (e.g., Nogueira et al., 2018; Saab et al., 2021). Therefore, flower size can be related to the general attractiveness of poricidal flowers to visitors and the patterns of flower visitation by animals differing in their capacity as pollinators.

This systematic review aimed to investigate the relationship between flower size and pollinator assemblage across plant species bearing poricidal flowers. We asked (i) how flower size influences the assemblage of bee species visiting the flowers of plant species bearing poricidal anthers; and (ii) how flower size is related to the size and vibrating behavior of the most frequent bee species visiting the flowers of different plant species with poricidal anthers. We hypothesized that large-flowered species are visited by more species than small-flowered species, leading to a more diverse visitor assemblage in species with large flowers, including both vibrating and non-vibrating species, pollinators and antagonists, respectively. We also hypothesized that large-flowered species are more likely to be visited by large-bodied bee species exhibiting a mandatory sonication than small-flowered species. To evaluate these hypotheses, we first conducted a systematic review of pollination studies describing the visitation patterns of flowers with poricidal anthers by different species. Second, we built a dataset describing variation in the flower size of plant species used as a model in the studies reviewed in our systematic search. Finally, we identified the most frequent bee species visiting the flowers in each study and extracted their body size from the literature.

2. Material and methods

2.1. Study case dataset

We built a dataset based on a systematic search of published studies describing the visiting patterns of pollen flowers with poricidal anthers in different plant species. Our systematic search utilized a combination of keywords that were used in different literature search engines: Web of Science (www.webofknowledge.com), Life Science Scopus (www.scopus.com), and Google Scholar (www.scholar.google.com) (Supplementary Material S1), we utilized different search engines to enhance our chances of locating relevant studies. The database included pollination studies available up to August 2021, and after removing duplicates, our sampling effort returned 300 potentially appropriate studies.

After our search, we screened the studies by reading each paper's abstract and methods. We included in our database the studies that (i) used as models noncommercial plant species bearing flowers with poricidal anthers, (ii) recorded the pattern of flower visitation over an explicit period (hours), and (iii) were conducted in the field. When we could not assess the first criterion directly from the study case, we retrieved the required information from taxonomic literature related to the plant species or from ecological literature about species with poricidal anthers (e.g., Buchmann, 1983; Vallejo-Marín et al., 2010; Vogel, 1978). After the screening, we got 32 suitable pollination studies, which summed data on 56 plant species with poricidal anthers. Two studies not initially recovered in our literature review were included. They are studies co-authored by the authors of this study (Delgado, 2019; Valadão-Mendes et al., 2022) and met our criteria. We surveyed the following information in each study for each poricidal flower species: scientific name and botanical classification at the family level, primary and secondary flower rewards, total observation time of flower visitors, and flower visitor species.

2.2. Flower size measurement

To measure the size of poricidal flowers of different plant species, we used digital images of each plant species available at online plant collection databases: speciesLink (http://inct.splink.org.br/), Reflora (https://floradobrasil.jbrj.gov.br), the New York Botanical Garden, NYBG (http://sweetgum.nybg.org); the Royal Botanic Gardens Kew (https://www.kew.org), and the Global Plants – JSTOR (https://plants.jstor.org). We considered that the images were suitable for our measurements when (i) the plant specimen had visible, well-preserved flowers, i.e., with parts not folded or damaged; (ii) an image with high resolution and available scale bar; (iii) flowers maintained structures attractive to a pollinator, i.e., corolla and androecium; (iv) plant specimen was identified by a specialist of its botanical group or could be otherwise identified by comparison to other identified specimens. We excluded duplicates and kept the images of 198 specimens representing 56 plant species (100% of model plant species used in the study cases). Specimen images were rare for some plant species, so the sampling size was variable across species (averaging 3.5 (range: 1–10) specimen images per species). However, it is essential to note that our focus is on describing the interspecific variation in floral size. This variation is expected to exceed intraspecific variation, meaning that any potential bias related to variation in our sampling size should play a minor role in our analysis.

To measure flower size of plant species included in our dataset, we used ImageJ 1.8.0. software (Schneider et al., 2012). For that, we selected five images among all the digital photos we retrieved for each plant species that met the selection criteria described above. Images that did not reach these criteria were discarded. Twenty-one plant species (37% of our dataset) had less than five suitable images for measuring. In those cases, we measured more than one flower per specimen image until we got five measurements of flower size for each plant species. In botanical collections, tridimensional floral structures are flattened into two dimensions after herborization, leading to distortion of the original flower shape. To avoid distortion bias, we measured flowers with non-twisted parts in each image (see our inclusion criteria above on the quality of flowers to be measured). These non-twisted flowers could still exhibit some distortion levels. However, as we were interested in descriptors of flower size rather than flower shape and the relative variation in these measurements across species, we considered that such distortion would have negligible effect on our measurements. Before each measurement, we calibrated digital photos based on the scale bar attached to vouchers (flower size measurement protocol - Supplementary Material S2). We used flower diameter as a proxy of flower size. We defined flower diameter as the linear distance between the two edges formed by the tip of the corolla at opposite sides of the flower (i.e., the corolla opening diameter), perpendicular to the flower's central axis (Supplementary Material S3). When flowers were tubular (e.g., Ericaceae species), we described flower size based on two linear measurements: diameter at the middle of the flower tube and diameter at the flower tube opening (Supplementary Material S3). We then utilized the diameter at the middle of the flower tube as a proxy of flower size for these tubular flowers. The diameter was larger at this level in the species of our dataset, representing the visual pattern of flower size and not constraints to the access of flower resources (e.g., diameter at the flower tube opening).

Based on flower diameter values, we classified flowers with poricidal anthers into three discrete categories: small flowers (diameter $\leq$ 0.99 mm), medium flowers (1.00 mm $\leq$ diameter $\leq$ 2.99 mm), and large flowers (diameter $\geq$ 3.00 mm). We applied this classification for two main reasons. First, the relationship between flower size and flower visitor diversity is expected to follow nonlinear trajectories (Kuppler et al., 2021; Olesen et al., 2007). In this case, we adopted categories using logarithmic classes of flower sizes across plant species, with higher categories including a larger range of flower size variation, to better represent the fundamental nature of the data and improve the fit of linear models. Second, flower measurements on dried specimens are likely to have some fraction of error compared to measurements on fresh materials, especially on larger flowers, but broad categories tend to decrease the bias related to such error. The three-level categorical predictor (small, medium, and large flowers) was used in all statistical tests and was called flower size throughout the study. Slight variations in the range of the three categories of flower size did not change the patterns detected in our study.

2.3. Flower visitors: species diversity, functional groups, and most frequent species

To evaluate the role of flower size on the functional diversity of bee assemblages visiting pollen flowers, we first classified flower visitors into ten groups (hereinafter termed visitor group) based on the functional group the visitor belongs to (Fenster et al., 2004; Koski et al., 2015; Moretti et al., 2009): beetle, ladybug, bedbug, spider, ant, wasp, fly, butterfly, and bees. Since bees are the main species performing buzz-pollination, we classified them into two distinct functional groups based on the genus-wide capacity of bee species in producing vibrations, according to Cardinal et al. (2018): (i) vibrating bee species and (ii) non-vibrating bee species. The capacity to produce flower vibrations is essential in these pollination systems because vibrating bee species are expected to extract pollen efficiently from poricidal anthers and ultimately pollinate flowers (Vallejo-Marín 2019). We assumed vibrating bee species as potential pollinators and non-vibrating bee species as antagonists or neutral visitors (e.g., Nogueira et al., 2018).

For each poricidal flower species, we counted the number of groups visiting the flowers; the number of visitor species that touched the stigma; the number of visitor species that sporadically or never touched the stigma; the number of bee species recorded on flowers; the number of vibrating bee species; the number of non-vibrating bee species; and the number of bee species that cut or chew anthers to extract pollen grains (e.g., *Trigona* species) mentioned in each study.

We also obtained detailed information about the most frequent visitor species for each plant species with poricidal flowers. We focused on the main flower visitor because the frequency of flower interaction with a given mutualistic partner is a reliable indicator of its net benefit for the plant partner (Vázquez et al., 2005). We accessed the frequency of flower visitors in most selected studies (60.6%). Still, in 37.8% of cases, we only had access to the species name of the most frequent flower visitor without any quantitative estimation of visitation frequency. In sum, we recovered the most frequent flower visitor in 98.4% of plant species analyzed, in which the main species visiting poricidal flowers was always a bee species. From the identity of the main bee species visiting each plant, we obtained data on the vibration behavior and body size for each case. First, we focused on the role of poricidal flower size on the probability of plants having as the main visitor vibrating bee species, i.e., species with vibrational behavior on poricidal anthers. We also obtained body length data of each study's most frequent bee visitor species. When the authors in each study did not explicitly report bee body length, we searched for such information in the taxonomic literature (Abrahamovich et al., 2005; Bezerra and Machado, 2003; Eltz et al., 2011; Fracasso and Sazima, 2004; Freitas and Silveira, 2017; Gaglianone, 2001; Gibbs, 2011; Liu and Koptur, 2003; Marchi and Alves-dos-Santos, 2013; Mawdsley, 2018; Nadia and Machado, 2005; Oliveira et al., 2008; Pereira et al., 2011; Sigrist and Sazima, 2015; Vivallo, 2019; Vivallo and Zanella, 2012). Moreover, we used the PaDiL database to consult taxonomic synonyms of the most frequent visitor bee species (https://www.padil.gov.au/pollinators/pest/main/138578). By doing so, we could retrieve body length data based on synonym descriptions or records. When body length data were described as a range (e.g., 9–14 mm), we used the mean between the maximum and minimum body length values reported for each bee species.

Since body length data were derived mainly from taxonomic descriptions and identification keys providing range statistics (37.7% of

the studies), we assumed a skeptical posture regarding body length values. Because of that, in addition to the evaluation of the role of flower size on mean body size of the main bee visitors, we also performed a categorical analysis investigating the role of flower size to assess the probability of poricidal flowers having a large-bodied bee species as the main flower visitor. In this case, we primarily adopted the classification of bee body size used in previous studies, grouping the bees into three categories: small bees with a body length < 10 mm, medium bees with a body length between ≤ 10 mm and ≤ 15 mm, and large bees with body length > 15 mm (Pereira et al., 2011; Stout, 2000). However, as a result of our relatively low sample size, we finally adopted only two categories to classify the most frequent visitor bee species: small-bodied (body length value < 15 mm) and large-bodied bee species (body length value ≥ 15 mm). We also considered body length continuously in the statistical analyses.

2.4. Statistical analysis

Across the 55 plant species of our dataset, three species offered pollen grains plus another reward to their floral visitors. Since the mechanisms driving the flower attractiveness to pollinators in these plants may differ from flowers offering only pollen (Parachnowitsch et al., 2019), we excluded these three species from our statistical analyses described below. Thus, we used information from 52 plant species offering only pollen to pollinators in our general analyses. Although unsuitable for formal statistical analysis, we also visually inspected the number of visitor species interacting with poricidal flowers in our complete dataset, including the three species with more than one flower resource (see Supplementary Material S4). The visual inspection can provide valuable descriptive information for future studies addressing this still-overlooked flower type.

We applied a generalized linear mixed model (GLMM) to test if flower size influences the characteristics of flower visitor assemblage, utilizing either a negative binomial or a Poisson distribution. We built one model for each of the following response variables: number of species visiting each flower species, number of groups of flower visitors, number of bee species visiting flowers, and number of vibrating and non-vibrating bee species visiting flowers. The predictor variable was always flower size described as a categorical variable with three levels: small, medium, and large flowers. We included the total period (hours) in which the flower visitors were observed in each study as an additional fixed factor and the study case as a random factor. The 'study case' factor sought to control the different methods applied in each study.

To evaluate if flower size influences mean body length of the most

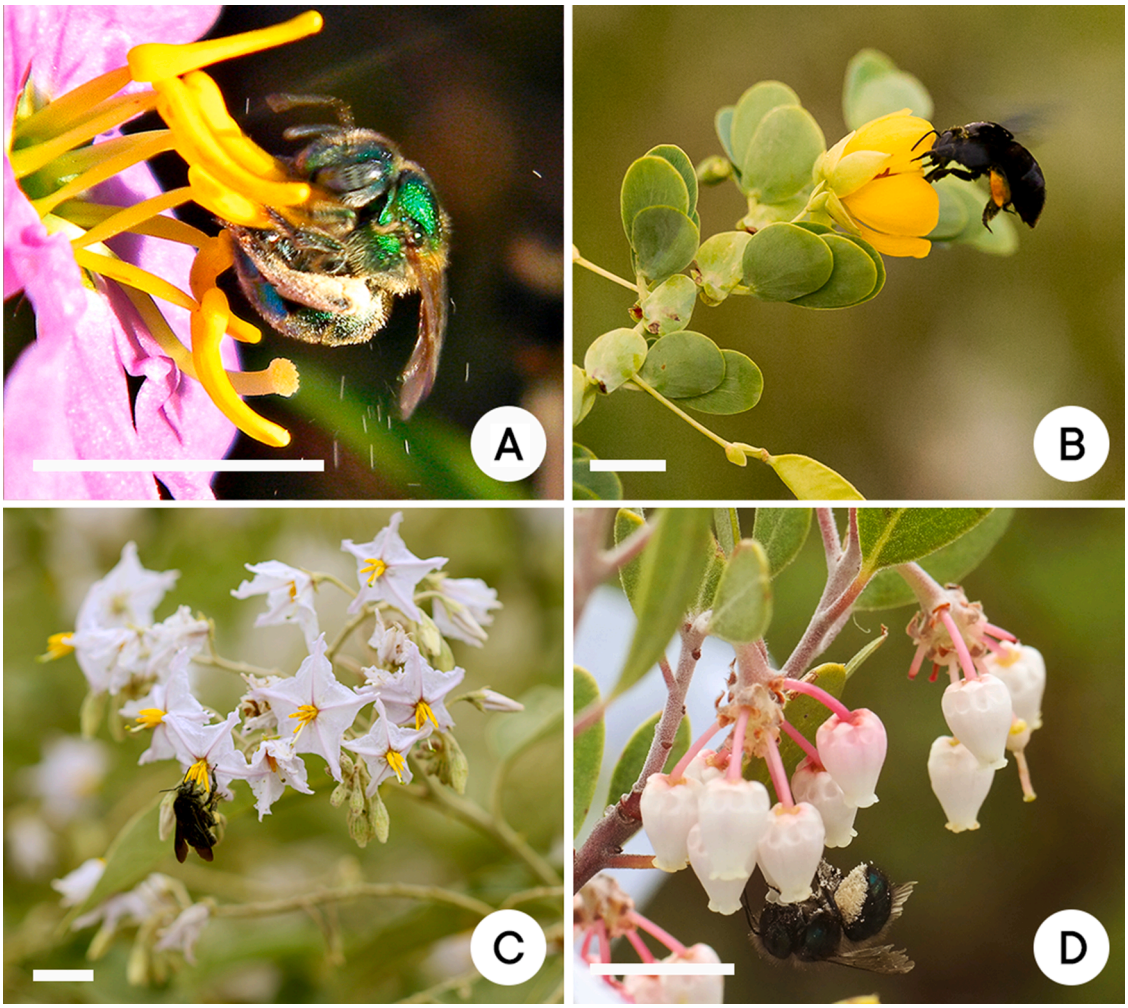


Fig. 1. Poricidal flowers visited by female bees collecting pollen grains. The first three images include representatives of the three families with the highest number of species available in our database. The fourth image represents a poricidal flower that also produces nectar. A – Poricidal flower of *Rhexia mariana* (Melastomataceae) receiving a visit of *Augochloropsis* sp. (Photo by Bob Peterson - https://www.flickr.com/photos/pondapple/10243478783/in/photostream/); B – Poricidal flower of *Chamaecrista latistipula* (Fabaceae) receiving a visit of *Eulaema nigrita* (Photo by Tamiris Delgado); C – Poricidal flower of *Solanum paniculatum* (Solanaceae) receiving a visit of *Bombus pauloensis* (Photo by César Augusto Arvelos) and D – Poricidal flower of *Arctostaphylos pungens* (Ericaceae) receiving a visit of *Andrena* sp. with white pollen grains accumulated in the bee's abdominal region (Photo by Anselmo Nogueira). Scale bars: 1 cm.

frequent bee species that visit the flower, we applied a generalized linear mixed model (GLMM) utilizing a negative binomial. Finally, we performed two binomial proportion tests to test if larger flowers are more likely to be visited by vibrating and large-bodied bees than smaller flowers. For these tests, we used the proportion of studies in which the main species visiting the poricidal flowers was a vibrating or large-bodied bee species. In each test, we used the size of the flowers (i.e., small, medium, or large) as the predictor variable.

All statistical analyses were conducted using the 'lme4' package (Bates et al., 2015) in the R environment version 4.1.2 (R Core Team, 2021).

3. Results

We sampled data on 55 plant species with poricidal flowers representing nine botanical families, summing 5124 h of observation of flower visitors. Most plant species with poricidal flowers belonged to three plant families: Melastomataceae (41.4%; Fig. 1A), Fabaceae (22.9%; Fig. 1B), and Solanaceae (14.3%; Fig. 1C). Plant species in our dataset mainly had medium flower sizes (57.1%), followed by large flower sizes (25.7%) and then small flower sizes (17.1%). See the complete list of poricidal flower species in Supplementary Material S5. The three species offering more than one reward to visitors included three families: Bixaceae, Ericaceae (Fig. 1D), and Krameriaceae. The species *Bixa orellana* and *Vaccinium uliginosum* ssp. *uliginosum* offered nectar in addition to pollen to its visitors, while *Krameria tomentosa* provided oil. In these studies, large flowers of *Bixa orellana* were visited by 16 species in 7 functional groups of visitors (beetle, ladybug, bed bug, ant, wasp, fly, and vibrating and non-vibrating bee species). In contrast, the tiny flowers of *Vaccinium uliginosum* ssp. *uliginosum* were visited by 29 species distributed in 4 functional groups of visitors (wasp, fly, butterfly, and vibrating and non-vibrating bee species). The species *Krameria tomentosa* was visited by 21 species in 2 groups of visitors (fly and bees). The most frequent visitor species across these three plant species were bee species.

3.1. Flower size and the assemblage richness of floral visitors

The floral size of pollen-flower species explained the number of species visiting flowers across plant species ($F = 4.85$; $p = 0.011$). Plant species with large flowers were visited by 2.4 times more species than those with small flowers (Fig. 2A). Bees were the only visitors of most plant species in our dataset (82%). The flowers were visited by bees and other visitor groups in 18% of plant species belonging to the following groups of visitors: beetles, flies, wasps, ants, and butterflies. Plant species were generally visited by only a few visitor groups (on average, one or two), regardless of flower size ($F = 1.78$; $p = 0.176$; Fig. 2B).

3.2. Flower size and the patterns of flower visiting by bees

Floral size explained variation across species in the number of bee species visiting flowers ($F = 4.09$; $p = 0.021$). The number of visiting bee species was 0.4 and 1.2 times greater among large-flowered plant species than medium-flowered and small-flowered plant species, respectively (Fig. 3). Most bee species (87.9%) visiting flowers with poricidal anthers exhibited vibrating behavior. Large-flowered plant species were visited two times more by vibrating bee species than small-flowered plant species ($F = 3.60$; $p = 0.033$; Fig. 4A). The number of vibrating bee species visiting large-flowered plant species was around 0.4 times greater than medium-flowered plant species and 1.1 times greater than small-flowered plant species. Floral size did not influence the number of non-vibrating bee species visiting flowers ($F = 1.87$; $p = 0.163$; Fig. 4B).

3.3. Flower size predicts the size and behavior of the most frequent visitor bee species

In all study cases, bees were the most frequent flower visitor species. We observed 35 species of bees acting as the main visiting species on flowers across plant species. The most frequent visitor species belonged to the genus *Bombus* (27.9%), *Xylocopa* (13.1%), *Augochloropsis* (13.1%), and *Centris* (11.5%).

The most frequent visiting bee species commonly had vibrating behavior (85.5% of cases). Large-flowered plant species were mainly visited by vibrating bee species (83.3% of cases; Fig. 5A). Similarly, vibrating bee species were those mainly visiting small- and medium-flowered plant species (87.8 and 91.7% of cases, respectively; Fig. 5A).

Large-bodied bee species were the main visitor in 77% of the studies included in our dataset (Fig. 5B); In contrast, medium- (60% of cases;

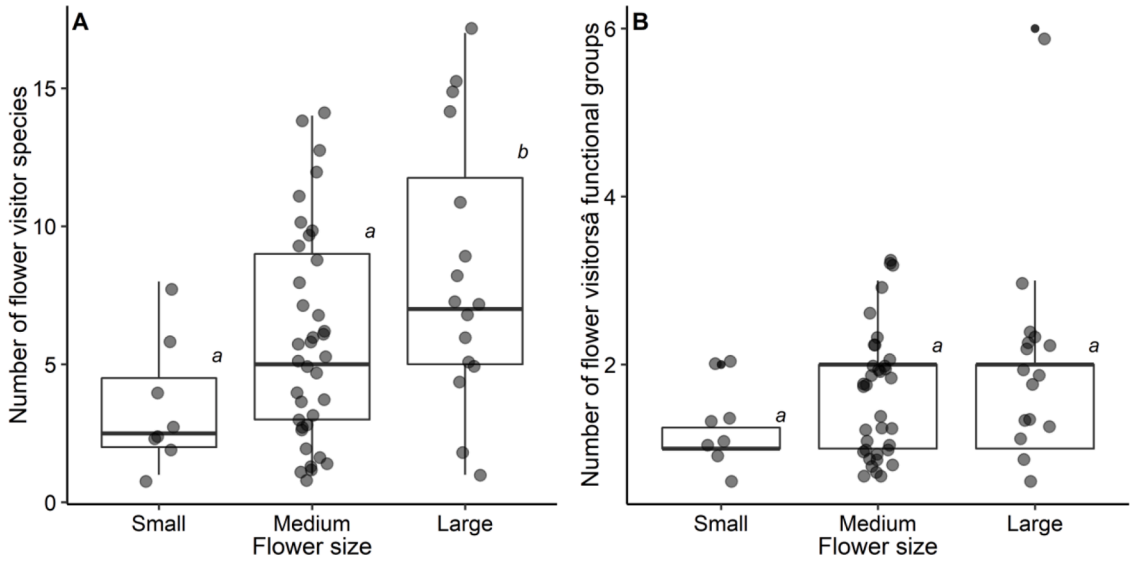


Fig. 2. The number of species and functional groups of floral visitors for different flower size categories. A) The number of floral visitor species recorded for different flower size categories; B) The number of floral visitor functional groups recorded for different flower size categories. Throughout, flower size categories reflect flower diameter values: 'small', flower diameter < 0.99 cm; 'medium', 1 cm $\leq$ flower diameter ≤ 2.99 cm; 'large', flower diameter > 3 cm. Boxes represent the 50% interquartile range, horizontal lines indicate the median, whiskers represent the 1.5*interquartile range, and points represent plant species recorded in the study case dataset. Different letters indicate significant differences among flower categories in the posthoc Tukey test ($p < 0.05$).

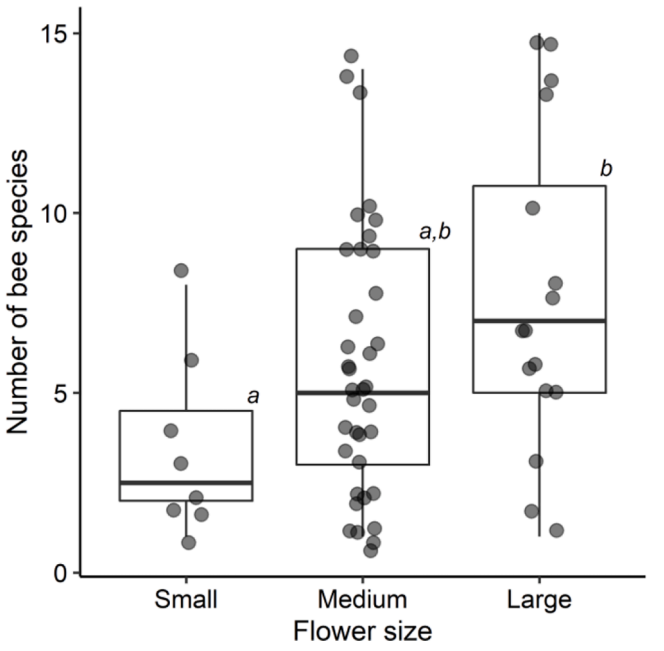


Fig. 3. The number of bee species visiting different flower size categories. Flower size categories reflect flower diameter values: 'small', flower diameter < 0.99 cm; 'medium', 1 cm ≤ flower diameter ≤ 2.99 cm; 'large', flower diameter > 3 cm. Boxes represent the 50% interquartile range, horizontal lines indicate the median, whiskers represent the 1.5*interquartile range, and points represent plant species recorded in the study case dataset. Different letters indicate significant differences among flower categories in the posthoc Tukey test ($p < 0.05$).

Fig. 5B); and small-flowered plant species (85% of cases; Fig. 5B) had a small-bodied bee species as the main visitor. In the quantitative evaluation, flower size predicted body length (mm) of the most frequent visitor bee species. The average size of the main bee species visiting the flowers was 14.4 mm ± 5.0. Bee species visiting large-flowered plant species were two times larger than those visiting small-flowered plant species ($F = 5.93$; $p = 0.004$; Fig. 6).

4. Discussion

We found that plant species with large poricidal flowers were visited by a larger and richer assemblage of floral visitor species than small-flowered plants. We also found that variation in flower size had different effects on the traits of visitor species. When looking at the whole assemblage of flower visitors, plant species with large and medium flowers interacted with both vibrating and non-vibrating bee species, while plant species with small flowers interacted almost exclusively with vibrating bee species. However, when focusing only on the most frequent visitor species, all flowers had a vibrating bee species as the main visitor, regardless of flower size. Still, large flowers were more likely to have a large-bodied bee species as the most frequent visitor than small flowers. The main species visiting large flowers had a body length two times larger than those visiting small flowers. Altogether, these results indicate that flower size affects the richness and functional diversity of the visitor's assemblage, as well as traits, mainly body size, of the main visitor species interacting with poricidal flowers.

As expected, large poricidal flowers were visited by a higher number of visitor species than medium and small flowers in our study. These results indicate that plant species with larger pollen-flowers can be more attractive to visitor species available in the habitat, especially bee species. In nectar flowers, the higher attractiveness of large flowers has been explained by the correlation between flower size and reward quantity offered to visitors (Totland and Matthews, 1998). Such correlation is expected to favor the cognitive association between flower size and nectar reward, driving visitors' preference towards larger nectar flowers (Kennedy and Elle, 2008). In the case of poricidal flowers, a body of evidence shows that flower size is related to the quantity of pollen offered to visitors (e.g., Solanaceae species - Vallejo-Marín et al., 2014). Also, floral size is correlated with other morphological traits, and such correlation could modify the flower's signaling to visitors. However, no evidence suggests that the spatial disposition of the reproductive structures can be cognitively associated with floral quality by floral visitors. Therefore, it is likely that the positive relationship between the size of poricidal flowers and the richness of visitors is explained by the same mechanisms driving this association in plant species with nectar flowers.

In our study, bee species were the most frequent visitors across plant

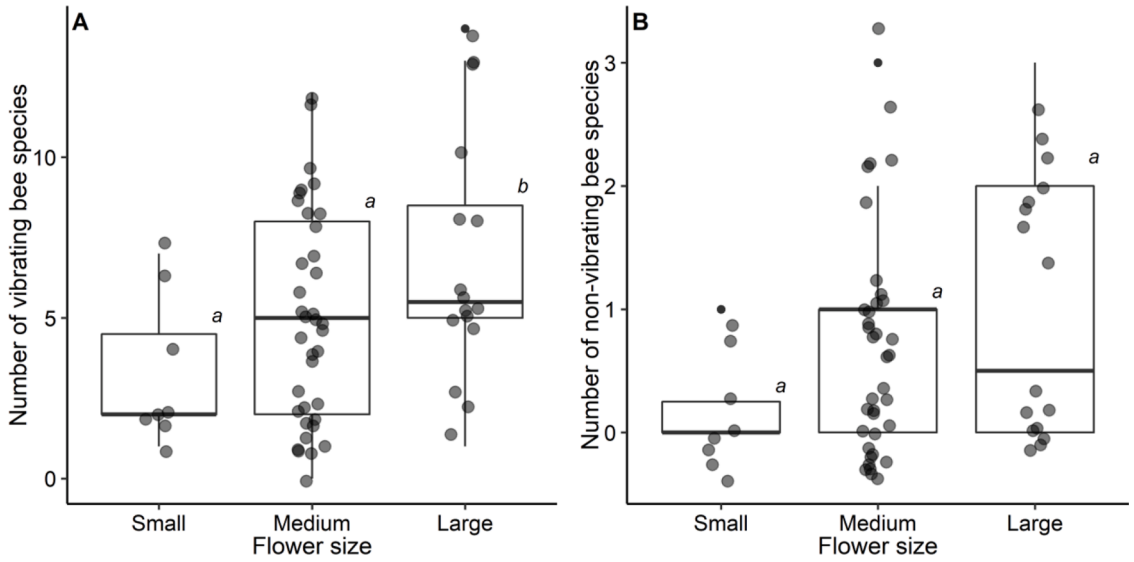


Fig. 4. The number of vibrating and non-vibrating bee species visiting different flower size categories. A) The number of vibrating bee species visiting different flower size categories; B) The number of non-vibrating bee species visiting different flower size categories. Throughout, flower size categories reflect flower diameter values: 'small', flower diameter < 0.99 cm; 'medium', 1 cm ≤ flower diameter ≤ 2.99 cm; 'large', flower diameter > 3 cm. Boxes represent the 50% interquartile range, horizontal lines indicate the median, whiskers represent the 1.5*interquartile range, and points represent plant species recorded in the study case dataset. Different letters indicate significant differences among flower categories in the posthoc Tukey test ($p < 0.05$).

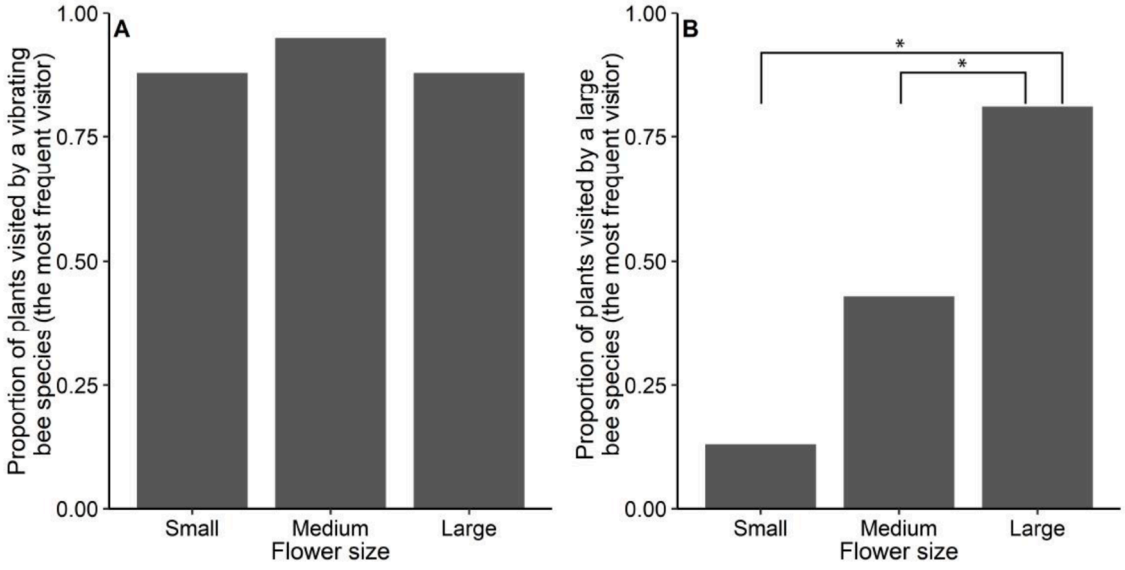


Fig. 5. Proportion of species visited by large-bodied and non-vibrating bee species into different flower size categories. A) Proportion of species visited by large-bodied bee species into different flower size categories. B) Proportion of species visited by non-vibrating bee species into different flower size categories. Throughout, flower size categories reflect flower diameter values: 'small', flower diameter < 0.99 cm; 'medium', 1 cm ≤ flower diameter ≤ 2.99 cm; 'large', flower diameter > 3 cm. Asterisks (*) indicate statistically significant differences ($p \leq 0.05$).

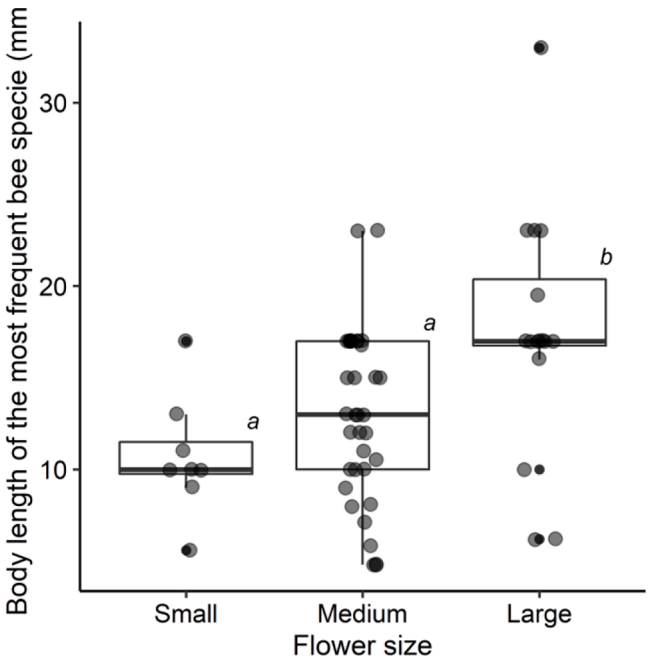


Fig. 6. Body length of the most frequent bee species visiting different flower size categories. Flower size categories reflect flower diameter values: 'small', flower diameter < 0.99 cm; 'medium', 1 cm ≤ flower diameter ≤ 2.99 cm; 'large', flower diameter > 3 cm. Boxes represent the 50% interquartile range, horizontal lines indicate the median, whiskers represent the 1.5*interquartile range, and points represent plant species recorded in the study case dataset. Different letters indicate significant differences among flower categories in the posthoc Tukey test ($p < 0.05$).

species with poricidal flowers, regardless of their size. This observation highlights the high ecological specialization and dependence between this flower type and bees. However, not all bee species interacting with poricidal flowers exhibit vibrating behavior (Mesquita-Neto et al., 2018). In our dataset, non-vibrating bee species were observed visiting medium and large flowers, while they were largely absent in species with small poricidal flowers. These non-vibrating bees may act as antagonists, removing pollen grains without performing pollination (Buchmann, 1983; Nogueira et al., 2018; Vallejo-Marín, 2019). In the absence of benefits, non-vibrating bees would impose pollen costs for plants depending on the time of interaction with the flowers (before or after the pollinator visit – Barker and Bronstein, 2016) and the quantity of pollen removed from the poricidal anthers. Also, non-vibrating bees exploit pollen with different behaviors, such as 1) "milking" the anther using their mandibles, plus buzzing (Roubik, 1989); 2) gleaning pollen leftovers that remain on flower organs after another bee visit (e.g., *Apis melifera* L.; Mesquita-Neto et al., 2018; Michener et al., 1978; Thorp and Estes, 1975, A. Nogueira pers. obs.); and 3) destroying the anthers with their mouth parts (Mesquita-Neto et al., 2018), such as Trigonini bees (e.g., *Trigona*). These behaviors (except the second one) were described across the studies reviewed here. While the first behavior potentially results in pollination, the second is unlikely to do so, as gleaning bees are considered commensals that do not impact reproductive plant success (Roubik, 1989). Differently, the third behavior is typically exploitative. Then, the positive association between flower size and reward quantity may make large flowers more attractive to pollinators and pollen larcenists.

We detected the existence of a morphological matching between flower size and body size of the most frequent bee species visitor across plant species. Plant species with larger flowers had a higher probability of interacting more frequently with large-bodied bee species, just as similarly found for small flowers and their small-bodied bee partners. The phenotypic matching between flowers and their visitors forecasts a greater rate of effective pollination (Whalen, 1978) since flower visitors are more likely to touch and deliver pollen grains to flower stigma when its body size has a better fit to floral morphology (Solís-Montero and Vallejo-Marín, 2017). However, the cause of phenotypic matching in poricidal plant species and their buzzing bee partners may not be as straightforward as in other pollination systems. Heavier bees from the same colony can have a greater vibration amplitude, increasing the amount of pollen ejected by the poricidal anthers (De Luca et al., 2013). Other studies evaluating insect vibration predicted this same pattern of vibrational amplitude by heavier bees (Cocroft et al., 2000; Cocroft and De Luca, 2005). Assuming that the bee size and weight are correlated (e.g., Corbet and Huang, 2014) in our dataset, it is expected that large-bodied bees will remove more pollen from large-anthered flowers

than small-bodied ones. Complementarily, phenotypic matching can also result from the morphological adjustment of the bee's body in relation to a flower's organs if we consider floral morphology of the poricidal flowers of the most representative plant families within our dataset – Fabaceae, Melastomataceae, and Solanaceae. Poricidal flowers of these plant groups usually have a suite of floral features, such as enantiostyly and hetheranthery (Buchmann, 1983), in which the proper size and positioning of pollinator in relation to the flower's reproductive organs (stigma and anthers) are important for successful pollination, especially in medium to large flowers where the distance of such organs is larger (Vallejo-Marín et al., 2014). These aspects deserve more attention in future studies exploring phenotypic matching of vibrating bee species and their pollen flower partners.

Since the most frequent partners are the main species that drive the net benefit in mutualistic interactions (Vázquez et al., 2005), it is possible that the benefits of having a high-quality matching partner as the main visitor offset the costs of facing a higher probability of exploitation by occasional non-vibrating visitors for plant species with larger poricidal flowers. Meanwhile, smaller poricidal flowers naturally face a lower chance of interacting with exploiters, while interacting more often with vibrating small-bodied bees that also phenotypically match their flower size. Therefore, interspecific variation in the size of poricidal flowers may reflect different reproductive strategies maximizing the reproductive success within different ecological contexts (Vallejo-Marín et al., 2014). In the case of small-flowered plants producing less pollen, the costs associated with pollen limitation should be offset by increased pollen transfer efficiency owing to the constraint of flower visitation by non-effective pollinators. In this case, legitimate visits of vibrating bees, morphologically coupled with the flower, would extract most of the pollen available in the flowers, ensuring both male and female plant reproductive success. Regarding large-flowered plants, the costs of being more attractive to ineffective pollinators can be offset by a higher probability of flower visitation by effective pollinators.

Our results on the patterns of flower visitation are described at the community level, reflecting the composition of the bee community occurring at each sampling site. More refined information about flower visitors, such as the flower visitation rates by different animal species, was largely absent. Although it does not invalidate our findings, investigations into the relationship between flower size and flower visitation rate by each visitor species should be approached in future studies and systematic reviews. Such research could provide a more accurate description of animal responses to flower traits, improving our ability to propose general mechanisms about the role of floral traits in attracting visitors to nectar and poricidal flowers. Another exciting avenue of investigation is to evaluate the influence of multiple flower rewards on the relationship between the size of poricidal flowers and patterns of flower visitation. In our systematic search, we found only three studies exploring poricidal species offering more than one reward type to its visitors (Carneiro et al., 2015; Mayer et al., 2012; Nascimento and Pereira, 2017). Since reward type is one trait that can also be cognitively associated with flower quality to visitors (Parachnowitsch et al., 2019), it is possible that the offering of multiple resources would lead to changes in the relationship between flower size and its relative value to visitors, consequently influencing the patterns of flower visitation by different species. For instance, the small-flowered plant species *Miconia brevitheca* and *Miconia tonduzii* var. *tonduzii* produce nectar and pollen rewards and are less commonly visited by vibrating bee species (Kriebel and Zumbado, 2014). Instead, the flowers of these species are often visited by syrphid and tachinid flies, which collect pollen and nectar from the flowers (Kriebel and Zumbado, 2014). This result is entirely different from the patterns we described here, in which bee species were the main flower visitor of species with poricidal flowers (93%).

In plants with poricidal flowers, variation in flower size is positively correlated with the distance between the anther and stigma floral organs (Vallejo-Marín et al., 2014). Therefore, small-flowered species could represent a modification of the plant's reproductive system, favoring the transference of pollen grains within flowers and decreasing plant pollinator dependence on reproduction ('selfing syndrome' – Sicard and Lenhard, 2011). However, the reduced separation of reproductive organs in small pollen-flowers is not consistently associated with changes in the poricidal nature of anthers, preventing pollen access to bee species exhibiting no vibrating behavior. This pattern could be changed only if small pollen-flowers, for instance, had a large anther pore opening, allowing non-vibrating bee species and other visitor types to be effective pollinators (Goldenberg et al., 2008). This is not the general pattern for species with small poricidal flowers included in our review. Some exceptions to this pattern were recorded among some Ericaceae that also offered nectar as a reward to its pollinating hummingbird species (de Araújo et al., 2011; Freitas, 2018; Luteyn, 2002; Navarro et al., 2008). In this plant group, different species with poricidal anthers with large anther pore openings also have large nectar disks near the flower ovary, thus increasing reward types and facilitating access to pollen grains, consequently increasing the functional diversity of flower visitors (Brito et al., 2016; Kriebel and Zumbado, 2014; Luteyn, 2002; Navarro et al., 2008).

5. Conclusions

Here, we showed that the size of poricidal flowers plays a significant role in shaping the size-traits bee species assemblages visiting these flowers and the behavioral traits of the most frequent species visiting these flowers. Despite such effect on flower visitation patterns, we advocate that the positive correlation between floral size and the amount of pollen offered by poricidal flowers leads to a situation in which the effect of flower size on the visitation patterns probably has a low impact on pollination outcome. Although larger flowers are more susceptible to interactions with exploiters attracted by the relatively large amount of pollen, reproductive success should be favored by the higher probability of having vibrating large-bodied bee species as the main visitor species; both traits are crucial for a high-quality pollinator. On the other hand, the reduction in flower size and the morphological and resource amount changes deriving from it probably prevent these flowers from interacting with non-vibrating bees, while favoring their visitation by vibrating small-bodied bee species as pollinators.

CRediT authorship contribution statement

Tamiris Delgado: Conceptualization, Data curation, Formal analysis, Methodology, Writing – original draft, Writing – review & editing. **Laura Carolina Leal:** Conceptualization, Supervision, Writing – review & editing. **Juliana Hanna Leite El Ottra:** Methodology, Writing – review & editing. **Vinicius Lourenço Garcia Brito:** Methodology, Writing – review & editing. **Anselmo Nogueira:** Conceptualization, Formal analysis, Funding acquisition, Methodology, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

We thank Bruna C. Barbosa and Guilherme V. Mota for help with the extraction of floral size and the rest of the LIPA (Laboratório de Interações Plant-Animal) for feedback and discussions during the

development of this project. Anonymous reviewers and Editor provided helpful comments on previous versions of this manuscript. This work is part of the doctoral thesis of T.D. This work was funded by a fellowship to T.D. from the Coordination for the Improvement of Higher Education Personnel (CAPES/grant number 88882.433263/2019-01) and by research grants to A.N. and V.L.G.B. of the Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq/grant numbers 434692/2018-2 and 308107/2021-7, respectively) and the São Paulo Research Foundation through a Young Investigators Grant (FAPESP/grant number 2019/19544-7).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.flora.2022.152198.

References

Abrahamovich, A.H., Díaz, N.B., Lucia, M., 2005. Las especies del género Bombus Latreille en Argentina (Hymenoptera: Apidae): estudio taxonómico y claves para su identificación. Neotropical Entomology 34, 235–250. https://doi.org/10.1590/S1519-566X2005000200012.

Armbruster, W.S., 2014. Floral specialization and angiosperm diversity: phenotypic divergence, fitness trade-offs and realized pollination accuracy. AoB Plants 6, 1–24. https://doi.org/10.1093/aobpla/plu003.

Ashman, T.L., Stanton, M., 1991. Seasonal variation in pollination dynamics of sexually dimorphic Sidalcea oregana ssp. spicata (Malvaceae). Ecology 72, 993–1003. https://doi.org/10.2307/1940599.

Bai, Y.P., Zhang, Y.W., Gituru, R.W., Zhao, J.M., Li, J.D., 2011. Sexual differences in reproductive characters and pollinator attractiveness in gynodioecious Glechoma longituba (Lamiaceae). Plant Species Biol. 26, 33–42. https://doi.org/10.1111/j.1442-1984.2010.00298.x.

Barker, J.L., Bronstein, J.L., 2016. Temporal Structure in Cooperative Interactions: What Does the Timing of Exploitation Tell Us about Its Cost? PLOS Biology 14, 1–16. https://doi.org/10.1371/journal.pbio.1002371.

Bates, D., Mächler, M., Bolker, B.M., Walker, S.C., 2015. Fitting linear mixed-effects models using lme4. Journal of Statistical Software 67, 1–48. https://doi.org/10.18637/JSS.V067.I01.

Bezerra, E.L.de S., Machado, I.C., 2003. Biologia floral e sistema de polinização de Solanum stramonifolium Jacq. (Solanaceae) em remanescente de mata atlântica, Pernambuco. Acta Botanica Brasilica 17, 247–257. https://doi.org/10.1590/S0102-33062003000200007.

Blarer, A., Keasar, T., Shmida, A., 2002. Possible mechanisms for the formation of flower size preferences by foraging bumblebees. Ethology 108, 341–351. https://doi.org/10.1046/j.1439-0310.2002.00778.x.

Brito, V.L.G., Sazima, M., 2012. Tibouchina pulchra (Melastomataceae): reproductive biology of a tree species at two sites of an elevational gradient in the Atlantic rainforest in Brazil. Plant Syst. Evol. 298, 1271–1279. https://doi.org/10.1007/s00606-012-0633-5.

Brito, V.L.G., Fendrich, T.G., Smidt, E.C., Varassin, I.G., Goldenberg, R., 2016. Shifts from specialised to generalised pollination systems in Miconieae (Melastomataceae) and their relation with anther morphology and seed number. Plant Biol. 18, 585–593. https://doi.org/10.1111/plb.12432.

Buchmann, S.L., 1983. Buzz pollination in angiosperms. In: Jones, C.E., Little, R.J. (Eds.), Miscellaneous: Handbook of experimental pollination biology, pp. 73–113.

Campbell, D.R., Waser, N.M., Price, M.V., Lynch, E.A., Mitchell, R.J., 1991. Components of phenotypic selection: pollen export and flower corolla width in Ipomopsis aggregata. Evolution 45, 1458–1467. https://doi.org/10.1111/j.1558-5646.1991.tb02648.x.

Cardinal, S., Buchmann, S.L., Russell, A.L., 2018. The evolution of floral sonication, a pollen foraging behavior used by bees (Anthophila). Evolution 72, 590–600. https://doi.org/10.1111/evo.13446.

Carneiro, L.T., Aguiar, A.J.C., Martins, C.F., Machado, I.C., Alves-dos-Santos, I., 2015. Krameria tomentosa oil flowers and their pollinators: bees specialized on trichome elaiophores exploit its epithelial oil glands. Flora Morphol. Distrib. Funct. Ecol. Plants 215, 1–8. https://doi.org/10.1016/j.flora.2015.06.002.

Chittka, L., Raine, N.E., 2006. Recognition of flowers by pollinators. Curr. Opin. Plant Biol. 9, 428–435. https://doi.org/10.1016/J.PBI.2006.05.002.

Cocroft, R.B., De Luca, P., 2005. Insect Sounds and Communication, Insect Sounds and Communication: Physiology, Behavior, Ecology, and Evolution. CRC Press.

Cocroft, R.B., Tieu, T.D., Hoy, R.R., Miles, R.N., 2000. Directionality in the mechanical response to substrate vibration in a treehopper (Hemiptera: Membracidae: Umbonia crassicornis). J. Comp. Physiol. 186, 695–705. https://doi.org/10.1007/s003590000123.

Corbet, S.A., Huang, S.Q., 2014. Buzz pollination in eight bumblebee-pollinated Pedicularis species: does it involve vibration-induced triboelectric charging of pollen grains? Ann. Bot. 114, 1665–1674. https://doi.org/10.1093/aob/mcu195.

de Araújo, F.P., Farias, Y.E., Oliveira, P.E., 2011. Biologia floral e visitantes de gaylussacia brasiliensis (spr.) meissner (ericaceae) - Uma espécie com anteras poricidas polinizada por beija-flores. Acta Bot. Bras. 25, 387–394. https://doi.org/10.1590/S0102-33062011000200015.

De Luca, P.A., Vallejo-Marín, M., 2013. What's the "buzz" about? The ecology and evolutionary significance of buzz-pollination. Curr. Opin. Plant Biol. 16, 429–435. https://doi.org/10.1016/j.pbi.2013.05.002.

De Luca, P.A., Bussière, L.F., Souto-Vilaros, D., Goulson, D., Mason, A.C., Vallejo-Marín, M., 2013. Variability in bumblebee pollination buzzes affects the quantity of pollen released from flowers. Oecologia 172, 805–816. https://doi.org/10.1007/s00442-012-2535-1.

Delgado, T., 2019. Como duas espécies de Chamaecrista lidam com o dilema do pélen? Variação no tamanho floral e na estratégia reprodutiva mediada pelas abelhas. Thesis. UNESP, SP, Brazil.

Diggle, P.K., 2014. Modularity and intra-floral integration in metameric organisms: plants are more than the sum of their parts. Philos. Trans. R. Soc. B Biol. Sci. 369, 1–8. https://doi.org/10.1098/rstb.2013.0253.

Eckhart, V.M., 1999. Sexual Dimorphism in Flowers and Inflorescences. In: Geber, M.A., Dawson, T.E., Delph, L.F. (Eds.), Gender and Sexual Dimorphism in Flowering Plants. Springer, Berlin, Heidelberg. https://doi.org/10.1007/978-3-662-03908-3_5.

Elle, E., Carney, R., 2003. Reproductive assurance varies with flower size in Collinsia parviflora (Scrophulariaceae). Am. J. Bot. 90, 888–896. https://doi.org/10.3732/ajb.90.6.888.

Eltz, T., Fritzsch, F., Pech, J.R., Zimmermann, Y., Ramírez, S.R., Quezada-Euan, J.J.G., Bembé, B., 2011. Characterization of the orchid bee Euglossa viridissima (Apidae: Euglossini) and a novel cryptic sibling species, by morphological, chemical, and genetic characters. Zool. J. Linn. Soc. 163, 1064–1076. https://doi.org/10.1111/j.1096-3642.2011.00740.x.

Fausto, J.A., Eckhart, V.M., Geber, M.A., 2001. Reproductive assurance and the evolutionary ecology of self-pollination in Clarkia xantiana (Onagraceae). Am. J. Bot. 88, 1794–1800. https://doi.org/10.2307/3558355.

Fenster, C.B., Armbruster, W.S., Wilson, P., Dudash, M.R., Thomson, J.D., 2004. Pollination syndromes and floral specialization. Annu. Rev. Ecol. Evol. Syst. 35, 375–403. https://doi.org/10.1146/annurev.ecolsys.34.011802.132347.

Fenster, C.B., Cheely, G., Dudash, M.R., Reynolds, R.J., 2006. Nectar reward and advertisement in hummingbird-pollinated Silene virginica (Caryophyllaceae). Am. J. Bot. 93, 1800–1807. https://doi.org/10.3732/ajb.93.12.1800.

Fidalgo, A.de O., Guimarães, D.M., Caldiron, G.T., Barbosa, J.M., 2018. Reproductive ecology of two pioneer legumes in a coastal plain degraded by sand mining. Hoehnea 45, 93–102. https://doi.org/10.1590/2236-8906-53/2017.

Figueiredo, P.S.de, Ribeiro, É.K.M.D., Lacerda, D.M.A., Girnos, E.C., 2009. Estratégia reprodutiva de Cochlospermum orinocense (Kunth) Steud.: fenologia, biologia floral e sistema de cruzamento em uma espécie pioneira de florestas na Amazônia. Rev. Bras. Bot. 32, 781–792. https://doi.org/10.1590/s0100-84042009000400016.

Fisbein, M., Venable, D.L.5, 1996. Evolution of Inflorescence Design: Theory and Data. Evolution 50, 2165–2177. https://doi.org/10.2307/2410688.

Fracasso, C.M., Sazima, M., 2004. Polinização de Cambessedesia hilariana (Kunth) DC. (Melastomataceae): sucesso reprodutivo versus diversidade, comportamento e freqüência de visitas de abelhas. Rev. Bras. Bot. 27, 797–804. https://doi.org/10.1590/S0100-84042004000400018.

Freitas, F.V., Silveira, F.A., 2017. Synopsis of the bee genus Thygater Holmberg 1884 (Hymenoptera, Apidae) in the Brazilian state of Minas Gerais, with the description of a new species and a key to all Brazilian species. Zootaxa 4238, 1–29. https://doi.org/10.11646/zootaxa.4238.1.1.

Freitas, L., 2018. We need to talk about floral larceny. Rodriguesia 69, 2223–2228. https://doi.org/10.1590/2175-7860201869446.

Gaglianone, M.C., 2001. Nidificação e forrageamento de Centris (Ptilotopus) scopipes Friese (Hymenoptera, Apidae). Rev. Bras. Zool. 18, 107–117. https://doi.org/10.1590/s0101-81752001000500008.

Gibbs, J., 2011. Revision of the metallic Lasioglossum (Dialictus) of eastern North America (Hymenoptera: Halictidae: Halictini). Zootaxa 3073, 1–216. https://doi.org/10.11646/zootaxa.3073.1.1.

Goldenberg, R., Penneys, D.S., Almeda, F., Judd, W.S., Michelangeli, F.A., 2008. Phylogeny of Miconia (Melastomataceae): patterns of stamen diversification in a megadiverse neotropical genus. Int. J. Plant Sci. 169, 963–979. https://doi.org/10.1086/589697.

Goulson, D., 1999. Are insects flower constant because they use search images to find flowers? Oikos 88, 547–552. https://doi.org/10.1034/j.1600-0706.2000.880311.x.

Gumbert, A., Kunze, J., Chittka, L., 1999. Floral colour diversity in plant communities, bee colour space and a null model. Proc. R. Soc. Lond. Ser. B Biol. Sci. 266, 1711–1716. https://doi.org/10.1098/RSPB.1999.0836.

Herrera, J., 2005. Flower size variation in Rosmarinus officinalis: individuals, populations and habitats. Ann. Bot. 95, 431. https://doi.org/10.1093/AOB/MCI041.

Hrncir, M., Gravel, A.-I.I., Schorkopf, D.L.P., Schmidt, V.M., Zucchi, R., Barth, F.G., 2008. Thoracic vibrations in stingless bees (Melipona seminigra):resonances of the thorax influence vibrations associated with flight but not those associated with sound production. J. Exp. Biol. 211, 678–685. https://doi.org/10.1242/jeb.013920.

Kennedy, B.F., Elle, E., 2008. The reproductive assurance benefit of selfing: importance of flower size and population size. Oecologia 155, 469–477. https://doi.org/10.1007/s00442-007-0924-7.

Koski, M.H., Meindl, G.A., Arceo-Gómez, G., Wolowski, M., LeCroy, K.A., Ashman, T.-.L., 2015. Plant–flower visitor networks in a serpentine metacommunity: assessing traits associated with keystone plant species. Arthropod Plant Interact. 9, 9–21. https://doi.org/10.1007/s11829-014-9353-9.

Kriebel, R., Zumbado, M.A., 2014. New reports of generalist insect visitation to flowers of species of Miconia (Miconieae: Melastomataceae) and their evolutionary implications. Brittonia 66, 396–404. https://doi.org/10.1007/s12228-014-9337-1.

Kuppler, J., Wieland, J., Junker, R.R., Ayasse, M., 2021. Drought-induced reduction in flower size and abundance correlates with reduced flower visits by bumble bees. AoB PLANTS 13, 1–76. https://doi.org/10.1093/aobpla/plab001.

Latty, T., Trueblood, J.S., 2020. How do insects choose flowers? A review of multi-attribute flower choice and decoy effects in flower-visiting insects. J. Anim. Ecol. 89, 2750–2762. https://doi.org/10.1111/1365-2656.13347.
Liu, H., Koptur, S., 2003. Breeding system and pollination of a narrowly endemic herb of the lower Florida keys: impacts of the urban-wildland interface. Am. J. Bot. 90, 1180–1187. https://doi.org/10.3732/ajb.90.8.1180.
Luteyn, J.L., 2002. Diversity, Adaptation, and Endemism in Neotropical Ericaceae : biogeographical Patterns in the Vaccinieae. Bot. Rev. 68, 55–87.https://www.jstor.org/stable/4354411.
Marchi, P., Alves-dos-Santos, I., 2013. As abelhas do gênero Xylocopa Latreille (Xylocopini, Apidae) do estado de São Paulo, Bras. Biota Neotrop. 13, 249–269. https://doi.org/10.1590/S1676-06032013000200025.
Mawdsley, J., 2018. New records and diagnostic notes on large carpenter bees (Hymenoptera: Apidae: Genus Xylocopa Latreille), from the Amazon River basin of South America. Insecta Mundi 631, 1–15. https://doi.org/10.5281/zenodo.3699701.
Mayer, C., Michez, D., Chyzy, A., Brédat, E., Jacquemart, A.L., 2012. The abundance and pollen foraging behavior of bumble bees in relation to population size of whortleberry (Vaccinium uliginosum). PLoS ONE 7, 1–10. https://doi.org/10.1371/journal.pone.0050353.
Mesquita-Neto, J.N., Blüthgen, N., Schlindwein, C., 2018. Flowers with poricidal anthers and their complex interaction networks—disentangling legitimate pollinators and illegitimate visitors. Funct. Ecol. 32, 2321–2332. https://doi.org/10.1111/1365-2435.13204.
Michener, C.D., Winston, M.L., Jander, R., 1978. Pollen manipulation and related activities and structures in bees of the family Apidae. Univ. Kansas Sci. Bull. 51, 575–601. https://doi.org/10.5962/bhl.part.17249.
Michener, C.D., 1962. An interesting method of pollen collecting by bees from flowers with tubular anthers. Rev. Biol. Trop. 10, 167–175.
Moretti, M., De Bello, F., Roberts, S.P.M., Potts, S.G., 2009. Taxonomical vs. functional responses of bee communities to fire in two contrasting climatic regions. J. Anim. Ecol. 78, 98–108. https://doi.org/10.1111/j.1365-2656.2008.01462.x.
Nadia, T.de L., Machado, I.C., 2005. Polinização por vibração e sistema reprodutivo de duas espécies de Sauvagesia L. (Ochnaceae). Rev. Bras. Bot. 28, 255–265. https://doi.org/10.1590/s0100-84042005000200006.
Nascimento, A., Pereira, L., 2017. Bixa orellana L. (Bixaceae): dependência de polinizadores e estratégias de forrageio dos visitantes florais. Biota Amaz. 7, 1–7. https://doi.org/10.18561/2179-5746/biotaamazonia.v7n3p1-7.
Navarro, L., Guitián, P., Ayensa, G., 2008. Pollination ecology of Disterigma stereophyllum (Ericaceae) in South-Western Colombia. Plant Biol. 10, 512–518. https://doi.org/10.1111/j.1438-8677.2008.00059.x.
Nogueira, A., Valadão-Mendes, L.B., El Ottra, J.H.L., Guimarães, E., Cardoso-Gustavson, P., Quinalha, M.M., Paulino, J.V., Rando, J.G., 2018. Relationship of floral morphology and development with the pattern of bee visitation in a species with pollen-flowers, Chamaecrista desvauxii (Fabaceae). Bot. J. Linn. Soc. 187, 137–156. https://doi.org/10.1093/botlinnean/boy008.
Olesen, J.M., Dupont, Y.L., Ehlers, B.K., Hansen, D.M., 2007. The openness of a flower and its number of flower-visitor species. Taxon 56, 737–756. https://doi.org/10.2307/25065856.
Oliveira, M.E.C., Poderoso, J.C.M., Ferreira, A.F., Lessa, A.C.V., Dantas, P.C., Ribeiro, G. T., Araújo, E.D., 2008. Análise melissopalinológica e estrutura de ninho de abelhas Trigona spinipes (Fabricius, 1793) (Hymenoptera: Apidae) encontradas no Campus da Universidade Federal de Sergipe, São Cristovão, SE. EntomoBrasilis 1, 17–22. https://doi.org/10.12741/ebrasilis.v1i2.14.
Parachnowitsch, A.L., Manson, J.S., Sletvold, N., 2019. Evolutionary ecology of nectar. Ann. Bot. 123, 247–261. https://doi.org/10.1093/aob/mcy132.
Pereira, A.C., da Silva, J.B., Goldenberg, R., Melo, G.A.R., Varassin, I.G., 2011. Flower color change accelerated by bee pollination in Tibouchina (Melastomataceae). Flora Morphol. Distrib. Funct. Ecol. Plants 206, 491–497. https://doi.org/10.1016/j.flora.2011.01.004.
Quakenbush, P.J., 2018. Pollination, Mating System, Phenology and Characterisation of Medinilla multiflora Merr. (Melastomataceae) on Mt Makiling, Philippines. Sibbaldia: the International Journal of Botanic Garden Horticulture 16, 121–139. https://doi.org/10.24823/sibbaldia.2018.251.
R Core Team (2022). R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. URL https://www.R-project.org/.
Rademker, M.C.J., De Jong, T.J., Klinkhamer, P.G.L., 1997. Pollen dynamics of bumble-bee visitation on Echium vulgare. Funct. Ecol. 11, 554–563. https://doi.org/10.1046/j.1365-2435.1997.00124.x.
Roubik, D., 1989. Ecology and Natural History of Tropical Bees. Cambridge University Press. https://doi.org/10.1017/CBO9780511574641.
Saab, G.S., Mansano, V.F., Nogueira, A., Maia, I.C., Bergamo, P.J., Paulino, J.V., 2021. A sophisticated case of division of labour in the trimorphic stamens of the Cassia fistula (Leguminosae) flower. AoB Plants 13, 1–17. https://doi.org/10.1093/aobpla/plab054.
Schneider, C.A., Rasband, W.S., Eliceiri, K.W., 2012. NIH Image to ImageJ: 25 years of image analysis. Nature Methods 9 (7), 671–675. https://doi.org/10.1038/nmeth.2089.
Sicard, A., Lenhard, M., 2011. The selfing syndrome: a model for studying the genetic and evolutionary basis of morphological adaptation in plants. Ann. Bot. 107, 1433–1443. https://doi.org/10.1093/aob/mcr023.
Sigrist, M.R.M.R., Sazima, M., 2015. Phenology, reproductive biology and diversity of buzzing bees of sympatric Dichorisandra species (Commelinaceae): breeding system and performance of pollinators. Plant Syst. Evol. 301, 1005–1015. https://doi.org/10.1007/s00606-014-1131-8.
Solís-Montero, L., Vallejo-Marín, M., 2017. Does the morphological fit between flowers and pollinators affect pollen deposition? An experimental test in a buzz-pollinated species with anther dimorphism. Ecol. Evol. 7, 2706–2715. https://doi.org/10.1002/ece3.2897.
Souza, I.M., Coutinho, K., Funch, L.S., 2012. Estratégias fenológicas de Senna cana (Nees & Mart.) H.S. Irwin & Barneby (Fabaceae: Caesalpinioideae) como mecanismo eficiente para atração de polinizadores. Acta Bot. Bras. 26, 435–443. https://doi.org/10.1590/s0102-33062012000200019.
Stout, J.C., 2000. Does size matter? Bumblebee behaviour and the pollination of Cytisus scoparius L. (Fabaceae). Apidologie (Celle) 31, 129–139. https://doi.org/10.1051/apido:2000111.
Thorp, R.W., Estes, J.R., 1975. Intrafloral behavior of bees on flowers of Cassia fasciculata. J. Soc. Entomol. Kansas 48, 175–184.
Totland, Ø., Matthews, I., 1998. Determinants of pollinator activity and flower preference in the early spring blooming Crocus vernus. Acta Oecol. 19, 155–165. https://doi.org/10.1016/S1146-609X(98)80019-2.
Vázquez, D.P., Morris, W.F., Jordano, P., 2005. Interaction frequency as a surrogate for the total effect of animal mutualists on plants. Ecol. Lett. 8, 1088–1094. https://doi.org/10.1111/j.1461-0248.2005.00810.x.
Valadão-Mendes, L.B., Rocha, I., Meireles, D.A.L., Leite, F.B., Sazima, M., Maruyama, P. K., Brito, V.L.G., 2022. Flower morphology and plant–bee pollinator interactions are related to stamen dimorphism in Melastomataceae. Plant Biol. 24, 240–248. https://doi.org/10.1111/plb.13359.
Vallejo-Marín, M., Manson, J.S., Thomson, J.D., Barrett, S.C.H., 2009. Division of labour within flowers: heteranthery, a floral strategy to reconcile contrasting pollen fates. J. Evol. Biol. 22, 828–839. https://doi.org/10.1111/j.1420-9101.2009.01693.x.
Vallejo-Marín, M., Walker, C., Friston-Reilly, P., Solís-Montero, L., Igic, B., 2014. Recurrent modification of floral morphology in heterantherous Solanum reveals a parallel shift in reproductive strategy. Philos. Trans. R. Soc. B Biol. Sci. 369, 1–11. https://doi.org/10.1098/rstb.2013.0256.
Vallejo-Marín, M., Da Silva, E.M., Sargent, R.D., Barrett, S.C.H., 2010. Trait correlates and functional significance of heteranthery in flowering plants. New Phytol. 188, 418–425. https://doi.org/10.1111/j.1469-8137.2010.03430.x.
Vallejo-Marín, M., 2019. Buzz pollination: studying bee vibrations on flowers. New Phytol. 224, 1068–1074. https://doi.org/10.1111/nph.15666.
Vivallo, F., Zanella, F.C.V., 2012. A new species of Centris (Paracentris) Cameron, 1903 from northeastern Brazil, with a key for the Centris species of the Caatinga region (Hymenoptera: Apidae). Zootaxa 16, 1–16. https://doi.org/10.11646/zootaxa.3298.1.1.
Vivallo, F., 2019. Taxonomic notes, lectotype designations and new synonymies in some species of the bee genus Centris Fabricius, 1804 described by Sándor Mocsáry and Heinrich Friese (Hymenoptera: Apidae). Zootaxa 4638, 534–546. https://doi.org/10.11646/zootaxa.4638.4.4.
Vogel, S., 1978. Evolutionary shifts from reward to deception in pollen flowers. In: Richards, A.J. (Ed.), The pollination of flowers by insects. Academic Press, London, pp. 89–96.
Whalen, M.D., 1978. Reproductive character displacement and floral diversity in solanum section androceras. Syst. Bot. 3, 77–86. https://doi.org/10.2307/2418533.